

NIRMA UNIVERSITY
School of Technology, Institute of Technology
B.Tech. CSE, Int. B. Tech. CSE-MBA
List of Practicals
Semester-V

Course Code 3CS101CC24	Subject Name Machine Learning	[3-0-2-4]
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Sr.	Title	Hours	CO
1	Introduction to Python, Numpy, Pandas, Matplotlib and Sklearn.	2	1,2
2	Use the pytesseract library in Python for optical character recognition from (i) an image file and (ii) a multi-page PDF file.	2	2,3
3	Simple and Multiple Linear Regression using Gradient Descent & Normal Equation Methods (without using sklearn or an equivalent library for both)	4	3,4
4	Linear Regression with Regularization (without using sklearn or equivalent library) and Simple and Multiple Linear Regression with and without regularisation using Sklearn	2	3,4
5	Implement Naïve-Bayes models (Multivariate Bernoulli, Multinomial and Gaussian using sklearn)	4	3,4
6	Implement Decision Trees (ID3, CART) using sklearn (2 Hrs). Compare the parameters such as criterion, min_samples_split, min_samples_leaf, max_depth, and their effects on accuracy.	2	3,4
7	Implement Support Vector Classification and Regression with Grid Search for Hyper-parameter tuning using sklearn. Plot the hyperplane. (3D plot)	4	3,4
8	Implement an AND gate using the Perceptron Learning Algorithm from scratch in Python using NumPy, Pandas, and Matplotlib. The use of built-in machine learning, neural network, or perceptron functions/libraries is not permitted.	4	3,4
9	Implement an EX-OR gate using the Perceptron Learning Algorithm from scratch in Python using NumPy, Pandas, and Matplotlib. The use of built-in machine learning, neural network, or perceptron functions/libraries is not permitted.	4	3,4
10	Implement Backpropagation Neural Network and K-means using sklearn.	2	3,4
	Total	30 Hours	